The Width of the Conformal Fan: Dependence and the Variance of Realized Coverage

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Abstract

Fix a split-conformal calibration set and ask how often a fresh, *independent* draw from the score marginal falls under the threshold $U_{(k)}$. That realized coverage $c = U_{(k)}$ is a random variable; for independent calibration scores it follows the $\mathrm{Beta}(k, n-k+1)$ law, variance about $\alpha(1-\alpha)/n$. We call this across-sample spread of realized coverage its *dispersion*, and the family of laws the dependence sweeps out the *fan* for short; we study how that dependence reshapes it. The organizing identity is that c is a functional of the count process alone, so its variance is governed by the count's covariance kernel, and negatively associated scores have a less variable count (Lemma 1). Three results follow. A leading-order fan coefficient: in a Bahadur regime $\mathrm{Var}(c) = n^{-2}\,\mathrm{Var}(N(1-\alpha)) + o(1/n)$, so the average pairwise covariance of the sub-threshold indicators sets the leading constant. An exact aggregate bound: summed over all levels k, negative association never widens the fan, by a convex-order contraction of the count against a concave functional and majorization. And, for equicorrelated normals, the single level is exact and monotone in the correlation, $\mathrm{Var}(Z_{(k)}) = v_k + \rho(1-v_k)$, dispersively so via strong log-concavity and Tweedie's formula. We keep c (coverage against an independent test draw) distinct from conformal conditional coverage with an exchangeable test point: under dependence the *mean* of c is not pinned, and the positive-dependence decomposition we give is an exact identity that does not by itself order the fan against the independent one. The general single-level negative-association contraction is left as a conjecture, supported numerically. The governing sign is the one the companion note attaches to the finite de Finetti measure.

1 Setup

Split conformal returns a set with marginal coverage $1-\alpha$. Fix the calibration set; the coverage you then realize on fresh test points is a random variable, and it is not $1-\alpha$ on the nose. Write the scores on the probability-integral scale, $U_i = F(S_i)$. We assume throughout that the score marginal F is continuous, so each $U_i \sim \mathrm{Unif}(0,1)$; if F has atoms, read U_i as the randomised PIT $F(S_i^-) + V_i\,(F(S_i) - F(S_i^-))$ with $V_i \sim \mathrm{Unif}(0,1)$ independent, which restores exact uniformity. With $k = \lceil (n+1)(1-\alpha)\rceil$ the conformal threshold is the order statistic $U_{(k)}$, and the calibration-conditional coverage against an independent test draw from the marginal is

$$c = U_{(k)}. \tag{1}$$

For *independent* calibration scores this is the classical result (Vovk, 2012; Marques F., 2024): $c \sim \mathrm{Beta}(k, n-k+1)$, with

$$\mathbb{E}[c] = \frac{k}{n+1}, \qquad \mathrm{Var}(c) = \frac{k(n-k+1)}{(n+1)^2(n+2)} \;\approx\; \frac{\alpha(1-\alpha)}{n}. \tag{2}$$

This across-sample spread of realized coverage is its *dispersion*, the formal object of this note; the family of such laws swept out as the calibration dependence varies is the *fan*, for short (we use the vivid word freely, but it always means this dispersion). The per-dataset coverage scatters around $1-\alpha$ with variance of order $1/n$ (standard deviation of order $n^{-1/2}$). The question of this note is what the dependence among the calibration scores does to it.

Two cautions fix the object of study. First, $c = U_{(k)}$ is coverage against a test draw *independent* of the calibration set; it is not the conformal conditional coverage $\mathbb{P}(U_{n+1} \leq U_{(k)} \mid U_1, \dots, U_n)$ for a test point exchangeable with, and dependent on, the calibration scores, which need not equal $U_{(k)}$. The marginal conformal guarantee comes from the rank symmetry of U_{n+1} among the $n+1$ scores and is not what controls c. Second, the mean $\mathbb{E}[c] = k/(n+1)$ is an *independence* statement, not a consequence of uniform marginals or exchangeability: if $U_1 = \cdots = U_n = V \sim \mathrm{Unif}(0,1)$ then $U_{(k)} = V$ and $\mathbb{E}[c] = \frac{1}{2}$ for every k. So under dependence both the mean and the variance of c respond; this note studies the variance, and flags the mean where it matters. The companion notes treat the orthogonal, within-dataset axis (Cotton; Cotton); here every statement is about the across-dataset fluctuation of c. The results separate by type: Theorems 2 and 4 and Propositions 1–2 are finite-sample exact, Theorem 1 is asymptotic and conditional on a Bahadur representation, and the general per-level negative-association contraction (Conjecture 1) is open.

A third point fixes what kind of dependence is in scope, since it has to square with an independent test. What the analysis requires is only that the test draw be independent of the calibration scores; the calibration dependence is then necessarily *design-induced* — a property of how the calibration set is assembled, not of a data stream the test belongs to. Antithetic pairs $(V, 1-V)$, stratified and Latin-hypercube designs, or balanced subsampling from a pool all make $(U_1, \dots, U_n)$ negatively associated while keeping each U_i marginally uniform, and a deployed test point is exogenous, hence independent. Dependence inherited from a source the test shares — the next step of a time series, or the unsampled remainder of a without-replacement draw — instead couples the test to the calibration, so that $c \neq U_{(k)}$ and a different analysis applies; that case is out of scope here. The maximally negatively associated floor is read the same way: a deterministic grid laid down by design, not a draw whose leftover is the test. This is the standard split-conformal test assumption (a fresh, independent future point); the only non-standard ingredient is letting the *calibration* set carry an engineered dependence.

The whole problem passes through the count process $N(u) = \sum_{i=1}^{n} \mathbf{1}\{U_i \leq u\}$, because $U_{(k)} = \int_0^1 \mathbf{1}\{N(u) < k\}\, \mathrm{d}u$, so

$$\mathrm{Var}(c) = \int_0^1 \int_0^1 \mathrm{Cov}\left(\mathbf{1}\{N(u) < k\},\, \mathbf{1}\{N(v) < k\}\right) \mathrm{d}u\, \mathrm{d}v, \tag{3}$$

a functional of the law of the count process alone — equivalently, by Sklar's theorem, of the score *copula*, since the uniform marginal has factored out (we return to this in Remark 6). Two facts about N drive everything.

Lemma 1 (Count variance)**.** *If the scores are negatively associated (Joag-Dev and Proschan, 1983), then for every u,* $\mathrm{Var}(N(u)) \leq n\,u(1-u)$*, with equality iff the sub-threshold indicators are pairwise uncorrelated.*

Proof. The indicators $\mathbf{1}\{U_i \leq u\}$ are functions of single distinct coordinates, all nondecreasing in the same direction, so negative association is preserved and they are pairwise non-positively correlated. So $\mathrm{Var}(N(u)) = \sum_i \mathrm{Var} + \sum_{i \neq j} \mathrm{Cov} \leq \sum_i u(1-u) = n\,u(1-u)$. □

2 The fan coefficient

Let $p = 1 - \alpha$ and $b_i = \mathbf{1}\{U_i \le p\}$, so $N(p) = \sum_i b_i$. (We reserve ξ for the i.i.d. normals of Proposition 2.) The representation is on the uniform scale, where the quantile density is $1/f(p) = 1$, so the usual $1/f$ factor is absent.

Theorem 1 (Fan coefficient)**.** *Suppose F is continuous (or a randomised PIT is used), $k/n \to p \in (0,1)$, and the scores satisfy a Bahadur representation $U_{(k)} - p = p - N(p)/n + r_n$ with the remainder negligible in L^2 relative to $n^{-1}\operatorname{Var}(N(p))^{1/2}$. Then*

$$\operatorname{Var}(c) = \frac{1}{n^2}\operatorname{Var}(N(p)) + o\Big(\frac{\operatorname{Var}(N(p))}{n^2}\Big). \tag{4}$$

If moreover $\operatorname{Var}(N(p)) \asymp n$, writing it as $np(1-p) + n(n-1)\bar{c}$ with $\bar{c} = \frac{1}{n(n-1)}\sum_{i\ne j}\operatorname{Cov}(b_i, b_j)$,

$$\operatorname{Var}(c) = \frac{p(1-p)}{n}\, R\,(1 + o(1)), \qquad R = 1 + \frac{n-1}{p(1-p)}\,\bar{c}. \tag{5}$$

So the leading fan coefficient is the count variance at the operating quantile. Under independence $\bar{c} = 0$, $R = 1$, recovering (2); negative association makes $\bar{c} < 0$ and $R < 1$, narrowing the fan. Two caveats the statement respects. The multiplicative form needs $\operatorname{Var}(N(p)) \asymp n$: under strong negative dependence the count variance can be $o(n)$ (zero at the floor), and then only the additive form is safe. And the Bahadur regime itself fails under a fixed positive latent (de Finetti) dependence, where $N(p)/n$ has a non-degenerate random limit $Q([0,p])$ and $U_{(k)}$ converges to a random conditional quantile rather than linearizing at p; the next section handles that regime exactly and separately.

3 Positive dependence: an exact decomposition

When the scores are extendable, drawn from an infinitely exchangeable sequence, de Finetti (de Finetti, 1937; Kerns and Székely, 2006) makes them conditionally independent given a latent directing measure Q. The law of total variance then gives an exact decomposition.

Theorem 2 (Positive-dependence decomposition)**.** *For an extendable exchangeable calibration sequence with directing measure Q,*

$$\operatorname{Var}(c) = \underbrace{\mathbb{E}_Q\big[\operatorname{Var}(U_{(k)} \mid Q)\big]}_{\textit{average within-Q order-stat variance}} + \underbrace{\operatorname{Var}_Q\big(m_k(Q)\big)}_{\textit{between-Q term}}, \qquad m_k(Q) = \mathbb{E}[U_{(k)} \mid Q]. \tag{6}$$

The between-Q term is non-negative and vanishes under independence.

The decomposition is exact and needs no asymptotics, but note what conditioning gives: $U_i \mid Q$ are i.i.d. Q, so $U_{(k)} \mid Q$ is the k-th order statistic of Q on the marginal scale, *not* a uniform Beta variable, and its mean $m_k(Q)$ moves with Q. The within-Q term is thus not the independent fan (2), and the identity by itself does *not* prove $\operatorname{Var}(c)$ exceeds the independent value: that comparison would need a separate theorem relating the average within-Q order-stat variance to the uniform fan. What the identity does isolate is the between-Q term, one signature of positive dependence (it also shifts the within-Q law from uniform to Q): a per-dataset latent shifts the whole calibration sample, and the realized coverage rides with it. Empirically the net effect is a wider fan, e.g. a one-factor Gaussian copula at correlation 0.2 gives $\operatorname{Var}(c) \approx 3.3$ times the independent value (and the comonotone limit $U_1 = \cdots = U_n$ gives $\frac{1}{12}$, far above it), but that widening is observed, not proved by Theorem 2 alone.

4 Negative dependence narrows the fan

The negative regime is the one a genuine prior cannot reach. By Kerns and Székely (2006) a finite exchangeable sequence with $\rho < 0$ has a *signed* directing measure, so the clean conditioning of Theorem 2 is unavailable: there is no Q to condition on, and the between-dataset term would be a negative quasi-variance. That is precisely the regime where the fan contracts.

Theorem 3 (Endpoints)**.** *Independent scores give* $c \sim \mathrm{Beta}(k, n-k+1)$*, the benchmark fan. At the exchangeable pairwise-correlation floor, take the discrete rank idealization in which the calibration scores are a uniformly random permutation of the grid* $\{1/(n+1), \dots, n/(n+1)\}$ *(sampling without replacement, pairwise correlation* $-1/(n-1)$*, the smallest attainable for an exchangeable vector): the order statistic is deterministic,* $U_{(k)} = k/(n+1)$ *a.s., so* $\mathrm{Var}(c) = 0$*.*

Independence is the benchmark, not the maximum: positive dependence can widen the fan well past it (the comonotone limit reaches $\frac{1}{12}$). The claim of this note is the other direction, that *negative* association does not widen it, with independence the (conjectural) upper endpoint of the negative-dependence range. Two model cautions on the floor. The grid permutation has *discrete*-uniform marginals, not the continuous $\mathrm{Unif}(0,1)$ of $U_i = F(S_i)$, so it sits in the model only if one allows discrete or rank scores; the continuous-marginal analogue is a Latin hypercube (one point per stratum, randomly permuted), whose fan is small, of order $1/n^2$, but not exactly zero. And the endpoints agree with Theorem 1 at $R = 1$ and $R = 0$. The two are joined by an exact path.

Proposition 1 (An exact variance interpolation)**.** *Let the calibration law be, with probability* $1-t$*,* n *independent uniforms, and with probability* t*, a contest (a uniformly random permutation of the grid* $\{1/(n+1), \dots, n/(n+1)\}$*). Both components yield realized coverage with the same mean* $k/(n+1)$*, the contest with zero variance, so by the law of total variance the between-component term vanishes and*

$$\mathrm{Var}(c) = (1-t)\,\frac{k(n-k+1)}{(n+1)^2(n+2)}. \tag{7}$$

Proof. Condition on the component. $\mathbb{E}[c \mid \mathrm{iid}] = \mathbb{E}[c \mid \mathrm{contest}] = k/(n+1)$, so $\mathrm{Var}(\mathbb{E}[c \mid \cdot]) = 0$, and $\mathbb{E}[\mathrm{Var}(c \mid \cdot)] = (1-t)\,\frac{k(n-k+1)}{(n+1)^2(n+2)}$. $\square$

So the fan collapses linearly to zero as $t \to 1$. Two honest caveats. The mixture's pairwise correlation is $\mathrm{Corr}(U_i, U_j) = -t/(n+1-2t)$, not $-t/(n-1)$: the grid component contributes $\mathrm{Cov} = -1/(12(n+1))$ but mixing changes the marginal variance to $(n+1-2t)/(12(n+1))$; the correlation reaches the floor $-1/(n-1)$ only at $t = 1$. And the mixture's one-dimensional marginal is neither continuous uniform nor discrete-grid uniform, so this is an exact *variance* interpolation, not a fixed-marginal copula path.

So we can prove the contraction exactly at both endpoints and along this whole interpolating family, and to leading order everywhere (Theorem 1). For an *arbitrary* negatively associated law the contraction also holds once it is summed over all levels.

Theorem 4 (Aggregate fan bound)**.** *For negatively associated scores with uniform marginals,*

$$\sum_{k=1}^{n} \mathrm{Var}(U_{(k)}) \;\le\; \sum_{k=1}^{n} \frac{k(n-k+1)}{(n+1)^2(n+2)}, \tag{8}$$

the independent value, with equality holding under independence. Summed over all target levels, the fan is never widened by negative association.

Proof. The order statistics are a permutation of the sample, so $\sum_k U_{(k)}^2 = \sum_i U_i^2$ and, under uniform marginals, $\sum_k \mathbb{E}[U_{(k)}^2] = n\,\mathbb{E}[U_1^2] = n/3$ for every such law; hence $\sum_k \mathrm{Var}(U_{(k)}) = \frac{n}{3} - \sum_k (\mathbb{E}\,U_{(k)})^2$. For the top j statistics, $\sum_{k>n-j} U_{(k)} = \int_0^1 \min\bigl(j, n-N(u)\bigr)\,\mathrm{d}u$. The function $\phi(m) = \max(-j, m-n)$ is convex and *nondecreasing*, which is exactly the class for which Shao (2000) compares a negatively associated sum to its independent counterpart, $\mathbb{E}\phi(N(u)) \le \mathbb{E}\phi(\mathrm{Bin}(n,u))$. (Because the means agree, $\mathbb{E}N(u) = nu = \mathbb{E}\,\mathrm{Bin}$, one may add any affine term, so the comparison in fact extends to all convex ϕ, the full convex order; the monotone-convex form already suffices here.) Since $\min(j, n-m) = -\phi(m)$, this gives $\mathbb{E}\min(j, n-N(u)) \ge \mathbb{E}\min(j, n-\mathrm{Bin}(n,u))$; integrating,

$$\sum_{k>n-j} \mathbb{E}\,U_{(k)} \;\ge\; \sum_{k>n-j} \tfrac{k}{n+1} \qquad \text{for every } j = 1, \ldots, n, \tag{9}$$

the right side being the independent profile's top-j sum. Both profiles are increasing and have the same total, $\sum_k \mathbb{E}\,U_{(k)} = \sum_i \mathbb{E}\,U_i = n/2 = \sum_k \frac{k}{n+1}$ (the case $j = n$ of (9), with equality). For two increasing vectors with equal sum, "every top-j partial sum of one dominates that of the other" is exactly the definition of majorization; so $(\mathbb{E}\,U_{(1)}, \ldots, \mathbb{E}\,U_{(n)})$ majorizes $(\frac{1}{n+1}, \ldots, \frac{n}{n+1})$. The map $x \mapsto \sum_k x_k^2$ is Schur-convex, hence larger on the majorizing vector, $\sum_k (\mathbb{E}\,U_{(k)})^2 \ge \sum_k (\frac{k}{n+1})^2$, and the displayed identity for $\sum_k \mathrm{Var}(U_{(k)})$ finishes it. □

The proof uses only identical uniform marginals and the convex-order domination $N(u) \le_{\mathrm{cx}} \mathrm{Bin}(n,u)$, not exchangeability as such. Equality holds under independence; that equality *forces* independence is not established here and would need a separate tightness lemma.

5 The single level: partial progress and a conjecture

Everything above is proved for general negative association. The single conformal level — the operationally relevant one — is not, and the obstruction is exactly why the aggregate succeeds: the step $\mathbf{1}\{N(u) < k\}$ behind a single $\mathrm{Var}(U_{(k)})$ is neither concave nor convex, so the convex order does not sign it, whereas the smooth aggregate functional $\min(j, n-N)$ is concave and it does. We collect here what *is* established at this level — exactly on the Gaussian scale, and reduced to a copula-diagonal inequality at the extreme $k = n$ — and state the general claim as a conjecture.

Conjecture 1 (Per-level contraction)**.** *If the calibration scores are negatively associated then* $\mathrm{Var}(c)$ *is at most the independent value* (2) *for each k individually.*

In the canonical continuous family the per-level statement is exact, and more, it is monotone in the correlation.

Proposition 2 (Exact per-level law for equicorrelated normals)**.** *Let $(Z_1, \ldots, Z_n)$ be equicorrelated standard normal with correlation $\rho \in [-\frac{1}{n-1}, 1]$, and let $v_k = \mathrm{Var}(\xi_{(k)})$ be the variance of the k-th order statistic of n i.i.d. standard normals. Then, for every k,*

$$\mathrm{Var}(Z_{(k)}) = v_k + \rho\,(1 - v_k). \tag{10}$$

As $v_k < 1$ for $n \ge 2$, this is strictly increasing in ρ: negative correlation strictly shrinks the variance of every order statistic, positive correlation inflates it.

Proof. Write $Z_i = \bar{Z} + (Z_i - \bar{Z})$. For equicorrelated normals $\bar{Z} \sim N(0, s)$, $s = (1 + (n-1)\rho)/n$, is independent of the centred vector, and $\mathrm{Cov}(Z_i - \bar{Z}, Z_j - \bar{Z}) = (1-\rho)(\delta_{ij} - 1/n)$, exactly $1 - \rho$ times the covariance of $(\xi_i - \bar{\xi})$ for i.i.d. standard normals. Hence $Z_{(k)} \overset{d}{=} \sqrt{s}\,\eta + \sqrt{1-\rho}\,(\xi - \bar{\xi})_{(k)}$

with $\eta \sim N(0,1)$ independent of the second term, so $\operatorname{Var}(Z_{(k)}) = s + (1-\rho)\operatorname{Var}\left((\xi - \bar{\xi})_{(k)}\right)$. Both terms are affine in ρ; matching v_k at $\rho = 0$ forces $\operatorname{Var}((\xi - \bar{\xi})_{(k)}) = v_k - 1/n$, and substituting gives $v_k + \rho(1 - v_k)$. ☐

Remark 1. *This is the per-level contraction, exact and monotone, on the Gaussian scale, unifying the two sides: the single line $v_k + \rho(1-v_k)$ runs from inflation at $\rho > 0$ through the independent value at $\rho = 0$ to the floor at $\rho = -1/(n-1)$. Under the Gaussian copula $U_i = \Phi(Z_i)$, which restores uniform marginals, $\operatorname{Var}(U_{(k)})$ is numerically monotone in ρ and below the independent value throughout $\rho < 0$ (`check_fan.py`). The transfer to the uniform scale is a dispersive-order statement, and it is half rigorous. Decompose along the common mean, $Z_{(k)} = \sqrt{1-\rho}\, D + \bar{Z}$ with $D = (\xi - \bar{\xi})_{(k)}$ and $\bar{Z} \sim N(0, s)$ independent. D is log-concave, being the order statistic of a vector with log-concave (Gaussian) joint density (Prékopa, 1973); and convolution with a log-concave density increases the dispersive order (Lewis and Thompson, 1981), so adding the Gaussian shift $\bar{Z}$ disperses, monotonically in s. The residual is that this outweighs the simultaneous $\sqrt{1-\rho}$ shrinkage of D as ρ grows, a one-parameter density-quantile monotonicity under the variance budget $\frac{n-1}{n}(1-\rho) + s = 1$, which no off-the-shelf convolution theorem delivers but which we now settle.*

Tweedie's formula sharpens that residual to a single posterior-variance bound. Reading $Y := Z_{(k)} = aD + b\eta$ ($a = \sqrt{1-\rho}$, $b = \sqrt{s}$) as Gaussian signal extraction of the log-concave signal D, the slope of the posterior mean equals the scaled posterior variance, $\frac{d}{dq}\mathbb{E}[D \mid Y = q] = a \operatorname{Var}(D \mid Y = q)/b^2$ (Efron, 2011; Hansen and Tong, 2026). The dispersive monotonicity is equivalent to this slope being at most $(n-1)a/n$, so, using $\frac{n-1}{n}a^2 + b^2 = 1$, the entire transfer reduces to

$$\operatorname{Var}(D \mid Y = q) \;\leq\; \frac{n-1}{n}\, b^2 \qquad \textit{for all } q, \tag{11}$$

a posterior-variance bound for log-concave Gaussian denoising, which the next lemma settles. The score-driven reading of the same identity (Hansen and Tong, 2026) is the econometric form of modelling the conditional spread, which is exactly what closes the information gap a conformal wrapper leaves (Cotton).

Lemma 2 (Strong log-concavity of D)**.** *$D = (\xi - \bar{\xi})_{(k)}$ is $\frac{n}{n-1}$-strongly log-concave, $(\log g_D)'' \leq -n/(n-1)$.*

Proof. The centred vector $w = \xi - \bar{\xi}$ has covariance $I - \frac{1}{n}J$, an orthogonal projection, so on the hyperplane $\{\sum_i w_i = 0\}$ it is a *standard* Gaussian: its $-\log$ density is $\frac{1}{2}\|w\|^2$, with Hessian the identity. The law of the ordered vector is this density restricted to the convex cone $\{w_1 \leq \cdots \leq w_n\}$. The cone enters only through the indicator of a convex set, a log-concave factor; writing the restricted density as $\exp(-\frac{1}{2}w^\top(I - \frac{1}{n}J)^+ w - \iota_C(w))$ with ι_C the convex indicator of the cone, the Hessian of the potential is $(I - \frac{1}{n}J)^+ + \nabla^2 \iota_C \succeq (I - \frac{1}{n}J)^+$. So the strong-log-concavity constant is set by the Gaussian quadratic alone; the cone can only raise it, never lower it, and in particular does not interfere through its boundary. Strong log-concavity passes to marginals (Prékopa, 1973; Brascamp and Lieb, 1976; Saumard and Wellner, 2014), the marginal of a coordinate being strongly log-concave to the reciprocal of that coordinate's reference variance; for $w_{(k)}$ this is $\frac{n-1}{n}$ (the diagonal of $I - \frac{1}{n}J$), giving constant at least $\frac{n}{n-1}$ (the parameter bookkeeping is in the Appendix). ☐

Proposition 3 (Gaussian-scale dispersive monotonicity)**.** *For equicorrelated standard normals with $\rho \in (-\frac{1}{n-1}, 1)$, $Z_{(k)}$ is increasing in the dispersive order as ρ increases.*

Proof. For a smooth one-parameter family the quantile evolves by $\partial_\rho q_\rho(u) = \mathbb{E}[\dot{Y} \mid Y = q_\rho(u)]$ with $\dot{Y} = \partial_\rho Z_{(k)}$, and dispersive monotonicity is exactly $\partial_\rho q_\rho(u)$ increasing in u, i.e. increasing in q.

Writing $Y = aD + b\eta$ and using $a\,\mathbb{E}[D \mid Y] + b\,\mathbb{E}[\eta \mid Y] = Y$ together with the budget $\frac{n-1}{n}a^2 + b^2 = 1$, this reduces to $m'(q) \le (n-1)a/n$, where $m(q) = \mathbb{E}[D \mid Y = q]$ (the computation of $M'(q)$ is in the Appendix). Tweedie's identity for Gaussian signal extraction gives $m'(q) = a\operatorname{Var}(D \mid Y = q)/b^2$. The posterior of D given Y has $-\log$ density with Hessian $(-\log g_D)'' + a^2/b^2 \ge \frac{n}{n-1} + \frac{a^2}{b^2} = \frac{n}{(n-1)b^2}$, the last equality by the budget and Lemma 2; the Brascamp–Lieb variance inequality (Brascamp and Lieb, 1976) then gives $\operatorname{Var}(D \mid Y = q) \le (n-1)b^2/n$, hence $m'(q) \le (n-1)a/n$. Equivalently the quantile density $q'_\rho(u) = 1/f_\rho(q_\rho(u))$ increases in ρ at each level (the density at each quantile decreases), which is the dispersive monotonicity. At the open-interval ends the statement is read in the limit: $b \to 0$ at $\rho = -1/(n-1)$ (the degenerate floor, $Z_{(k)} \to \sqrt{n/(n-1)}\,D$) and $a \to 0$ at $\rho = 1$ ($Z_{(k)} \to \eta$). $\square$

This settles the Gaussian-scale per-level statement in the strong (dispersive, not merely variance) sense. The one step left for the uniform-marginal conjecture under the Gaussian copula is the passage through Φ: being S-shaped, Φ is neither convex nor concave, so it need not transmit the dispersive order, and $\operatorname{Var}(U_{(k)}) = \operatorname{Var}(\Phi(Z_{(k)}))$ monotone in ρ is confirmed numerically (`check_fan.py`) but not by a transfer theorem.

The Gaussian case fixes the copula and varies k; the opposite slice, the top level $k = n$ for *any* negatively associated sample, is also exactly tractable and is where the conjecture becomes one-dimensional. There $U_{(n)} \le u$ iff every score lies below u, so the law of $U_{(n)}$ is the copula diagonal $\delta(u) = C(u, \dots, u) = \mathbb{P}(N(u) = n)$, and $\operatorname{Var}(U_{(n)}) = \int_0^1\!\int_0^1 [\delta(u \wedge v) - \delta(u)\delta(v)]\,du\,dv$. Negative association enters only as the negative-orthant bound $\delta(u) \le u^n$ (Joag-Dev and Proschan, 1983) — equivalently the convex-order image of the discretely convex indicator $\mathbf{1}\{N = n\}$. Writing the deficit $g(u) = u^n - \delta(u) \ge 0$ and $\mu^* = n/(n+1) = \mathbb{E}[U^{\text{iid}}_{(n)}]$, the variance has the exact closed form

$$\operatorname{Var}(U_{(n)}) = \operatorname{Var}^{\text{iid}}(U_{(n)}) + 2\int_0^1 (u - \mu^*)\,g(u)\,du - \Big(\int_0^1 g\Big)^2, \tag{12}$$

so the per-level conjecture at $k = n$ is *equivalent* to a centroid inequality on the deficit,

$$\frac{\int_0^1 u\,g(u)\,du}{\int_0^1 g(u)\,du} \;\le\; \mu^* + \tfrac{1}{2}\int_0^1 g \;: \tag{13}$$

the mass by which the diagonal falls short of independence must sit, on average, at most a half-deficit above the independent-maximum mean. This is *not* a consequence of the diagonal alone. A linear program over the functions meeting every condition a copula diagonal must satisfy (monotone, $0 \to 1$, $\delta \le u$, n-Lipschitz, Fréchet lower bound; Nelsen, 2006) together with the negative-orthant bound $\delta \le u^n$ returns a diagonal that violates (13) (`check_fan.py`). The single level is therefore irreducibly a property of the off-diagonal negative-orthant probabilities across thresholds, not of the diagonal that determines $U_{(n)}$'s law: exactly the content the aggregate convex-order bound uses in bulk and that a one-level statement cannot read off. It is the same obstruction that blocks localizing Theorem 4 to a single k, now in closed form, and every negatively associated structure tested satisfies (13) with margin.

The same convex-order contraction $N(u) \le_{\text{cx}} \operatorname{Bin}(n, u)$ that drives the aggregate bound is available per level, but turning it into the single-k statement needs the tail $\mathbb{P}(N(u) \ge k)$ to cross the binomial tail once in u, which the total positivity of the binomial tail in (k, u) (Karlin, 1968) makes plausible, followed by a single-crossing-implies-less-spread step. We claim only the variance statement, and deliberately not the stronger dispersive ordering: an order statistic of a dependent sample need not be dispersively smaller than the independent one even at fixed marginals (Panja et al., 2022), so the result is specific to the variance and is not a corollary of a dispersive theorem. The natural sufficient condition on the dependence is negative supermodular dependence (Christofides and Vaggelatou, 2004), which all the structures below satisfy. The table is illustrative, each row a single Monte Carlo estimate (6×10^4 draws, so the variance entries carry a relative Monte Carlo error of a few percent), fully reproduced by `check_fan.py` ($n = 20$, $\alpha = 0.1$, independent fan variance 3.92×10^{-3}); it is evidence for the conjecture, not a proof.

calibration dependence	$\mathbb{E}[c]$	$\mathrm{Var}(c)$	ratio to fan	sign
independent ($\rho = 0$)	0.904	3.92×10^{-3}	1.00	—
exchangeable, $\rho = -1/(n-1)$	0.923	1.16×10^{-3}	0.29	neg. assoc.
MA(1), lag-1 corr $-\frac{1}{2}$	0.912	2.71×10^{-3}	0.69	neg. assoc.
random negatively-paired blocks	0.909	3.14×10^{-3}	0.80	neg. assoc.
contest / without replacement	0.905	0	0.00	floor
one-factor, $\rho = +0.2$	0.863	1.33×10^{-2}	3.41	positive

Every negatively associated structure, exchangeable or not, comes in under the independent fan; the positive control widens it. The mean of c also responds to dependence, drifting up under negative and down under positive (to 0.863 in the one-factor row), confirming it is not pinned; the variance is what we track here. A wider search agrees: over exchangeable, moving-average, block, and randomly-generated non-exchangeable negatively-correlated Gaussian copulas at $n \leq 8$ and every k, the ratio to the independent fan never exceeds one (it ranges from about 0.30 at the floor to 0.80 for weak dependence), with no counterexample. In all of these the order-statistic CDFs $\mathbb{P}(N(u) \geq k)$ cross the independent ones exactly once, the single crossing the remark below uses.

Remark 2 (A closed form, and what an exact proof must dominate). *The variance is an explicit functional of $g(u) = \mathbb{P}(N(u) \geq k)$,*

$$\mathrm{Var}(c) = 2\int_0^1 (1-u)\, g(u)\,\mathrm{d}u - \Big(\int_0^1 g(u)\,\mathrm{d}u\Big)^2, \tag{14}$$

so writing $\delta = g - g^{\mathrm{iid}}$, $A = \int_0^1 g^{\mathrm{iid}} = 1 - \mathbb{E}[c^{\mathrm{iid}}]$, and $\Delta = \int_0^1 \delta$,

$$\mathrm{Var}(c) - \mathrm{Var}(c^{\mathrm{iid}}) = 2\int_0^1 \big[(1-u) - A\big]\,\delta(u)\,\mathrm{d}u - \Delta^2. \tag{15}$$

The $-\Delta^2$ term is non-positive. The weight $(1-u) - A$ changes sign at $u = \mathbb{E}[c^{\mathrm{iid}}] = k/(n+1)$. Empirically (across every structure in the table, and in the wider search below) δ changes sign exactly once, near $u \approx k/n$; we emphasize this is an observed single-crossing, not a consequence of Lemma 1, since count-variance contraction alone does not force one-crossing of the tail curve g. Granting that single crossing, the two sign changes agree up to the conformal offset between k/n and $k/(n+1)$, a band of width $O(1/n)$ on which the integrand is positive and elsewhere negative, so the identity gives the contraction up to that $O(1/n)$ boundary term, a second route to the leading order of Theorem 1, and isolates what a finite-sample proof must control: the boundary remainder against the slack Δ^2.

Remark 3 (Why the closed form does not close it, and the lemma that would). *The closed form is, by itself, circular. Substituting the moment identities $\int u\,\delta = \frac{1}{2}(\mathbb{E}[c_{\mathrm{iid}}^2] - \mathbb{E}[c^2])$ and $\int \delta = \mathbb{E}[c_{\mathrm{iid}}] - \mathbb{E}[c]$ into any elementary rearrangement of the difference, single-crossing splits, Abel summation, the exact first-order perturbation, collapses it back to $\mathrm{Var}(c) - \mathrm{Var}(c_{\mathrm{iid}})$ with the auxiliary crossing point cancelling: these structural facts carry no magnitude. Nor is the per-level convex order of Lemma-1* *enough on its own. Because* $\mathrm{Var}(c)$ *depends only on the single curve* $g(u) = \mathbb{P}(N(u) \geq k)$*, one might hope* $N(u) \leq_{\mathrm{cx}} \mathrm{Bin}(n, u)$ *at each* u *suffices; it does not, since a curve dominated pointwise can have larger variance (place an atom at the boundary), and such curves are excluded only because they are not realizable by a genuine sample. The content is therefore an extremal statement about the realizable curves, not their marginals:*

> *Open lemma. Among all* $g(u) = \mathbb{P}(N(u) \geq k)$ *realizable by a negatively associated uniform sample, the independent (binomial) curve maximizes* $\mathrm{Var}(c)$*.*

This is what a total-positivity or coupling argument would have to deliver. Proposition 1 and the contest endpoint are the cases settled by hand.

Remark 4 (The signed measure as a negative between-term)**.** *Theorem 2 reads the fan as within-dataset variance plus a non-negative between-dataset variance. In the signed-de-Finetti picture of the companion note, the negative corner is exactly where that between-dataset object turns negative, a Feynman–Wigner quasi-variance that subtracts from the within-dataset fan rather than adding to it, reaching the full cancellation* $R = 0$ *at the contest floor. This is heuristic, since a signed measure has no honest conditional variance, but it is the right cartoon: the same sign that decides conditional adaptivity in Cotton decides whether the fan is inflated or cancelled here.*

6 A constructive corollary

If negative dependence narrows the fan, one can engineer it, and there is a clean continuous-marginal construction that stays inside the $U_i \sim \mathrm{Unif}(0, 1)$ model (unlike the discrete grid of Theorem 3). Take a one-per-stratum design: with π a uniform random permutation of $\{1, \ldots, n\}$,

$$U_i = \frac{\pi(i) - 1 + V_i}{n} \quad \text{(Latin hypercube, } V_i \overset{\text{iid}}{\sim} \mathrm{Unif}(0,1)\text{)}, \qquad \text{or} \qquad U_i = \frac{\pi(i) - 1 + T}{n} \quad \text{(common jitter, } T \sim \text{Unif} \tag{16}$$

Each U_i is exactly $\mathrm{Unif}(0, 1)$ and the sample is exchangeable, with one point per stratum, so the k-th order statistic lies in the k-th stratum and $\mathbb{E}[U_{(k)}] = (k - \frac{1}{2})/n$, $\mathrm{Var}(U_{(k)}) = 1/(12n^2)$, a fan of order $1/n^2$ rather than $1/n$. (Note these continuous designs give mean $(k - \frac{1}{2})/n$, not the grid's exact $k/(n + 1)$.) More generally the classical variance-reduction designs apply: stratification never increases variance (Cochran, 1977), antithetic pairing helps for the monotone empirical CDF, and determinantal or repulsive designs give negative association with faster-than-independent concentration (Hough et al., 2006; Bardenet and Hardy, 2020).

Remark 5 (What is and is not admissible)**.** *The marginal guarantee needs each calibration score exchangeable with the test score, not the calibration scores independent of one another, so negative association is in principle compatible. But the routes differ in what they cost. Score-dependent curation of the calibration set, the obvious way to balance it, breaks exchangeability with the test point and voids the guarantee. A pooling that uses all* $n + 1$ *scores together (pooled ranks or de-meaning) preserves rank validity, but it is transductive, full-conformal in flavour, not split conformal with a threshold fixed before the test point is seen, and once the test score enters the pooled construction the realized-coverage variable is no longer the fixed-threshold* $c = U_{(k)}$ *studied here. The safe message is the design one: a pre-specified, exchangeability-preserving randomized design can lower the run-to-run variance of the threshold without disturbing validity. What makes such a design admissible is not that it is merely randomized and pre-specified, but that it preserves the exchangeability the validity argument needs; covariate stratification qualifies only when it is external to the responses and applied symmetrically across the* $n + 1$ *points. Exact zero fan is* *the discrete rank/grid idealization; continuous-marginal designs (Latin hypercube) give a small but nonzero fan, of order $1/n^2$. The tension is intrinsic: a generic negatively associated design lowers the dispersion only by also nudging $\mathbb{E}[c]$ off $1-\alpha$ (the mean is not pinned), so coverage that is both reproducible and on-target demands a mean-preserving symmetric design. Distribution-free validity and minimal coverage dispersion cannot both be driven to their limit at once, save in the degenerate zero-fan grid. A deeper caveat: the dispersion is a within-model quantity, computed with the marginal F and the copula C held fixed. Shrinking it by design lowers sampling variability but does nothing for model risk — whether F is the right target at all — and removes the run-to-run scatter that is the cheapest diagnostic of misspecification. A vanishing fan against an assumed F certifies reproducibility, not correctness; the zero-fan grid reads $1-\alpha$ every draw by construction, not because the world obliges.*

Remark 6 (The copula viewpoint)**.** *Because the scores are read on the uniform PIT scale, the joint law of $(U_1,\dots,U_n)$ is, by Sklar's theorem, a copula C, with the marginal F factored out entirely. So $\mathrm{Var}(c)$ is a copula functional: the fan is a copula invariant, identical for any score whose calibration dependence is C. The fan coefficient is read off the bivariate copula diagonal, $\bar{c} = \frac{1}{n(n-1)}\sum_{i\neq j}\left(C_{ij}(p,p)-p^2\right)$ at the operating quantile $p=1-\alpha$; for the maximum ($k=n$) $\mathrm{Var}(c)$ is exactly a functional of the diagonal $u\mapsto C(u,u)$, since $\mathbb{P}(U_{(n)}\leq u) = C(u,u)$. The endpoints are the Fréchet bounds: independence Π gives the Beta fan, the comonotone copula M ($U_1=\cdots=U_n$) the maximal fan $\frac{1}{12}$; the negative side has no dual, as the countermonotone lower bound is not a copula for $n>2$, which is why the floor is only the most negative exchangeable copula, $\rho=-1/(n-1)$. Theorem 2 then says that the extendable copulas, the exchangeable Archimedean families (Clayton, Gumbel, Frank), each with a de Finetti frailty W as directing variable, carry a non-negative between-Q term and widen the fan; the negative regime is the signed-de-Finetti corner with no such mixture, which is why its contraction needed convex order rather than conditioning. Finally, tail dependence acts through the copula's tails: a copula dependent in one tail pins the order statistics there and fattens the opposite end, so lower-tail dependence (Clayton) inflates the high, conformal quantile more than upper-tail dependence (Gumbel) — the reverse of the naive guess, as the companion demonstration shows.*

7 Related work

The $\mathrm{Beta}(k,n-k+1)$ law of calibration-conditional coverage and its variance are Vovk (2012) and Marques F. (2024); to our knowledge the papers cited here do not study the dependence among calibration scores as a control variable for the realized-coverage variance. Negative association and its consequences are Joag-Dev and Proschan (1983); that negatively associated sums are dominated in convex order by independent ones is Shao (2000), and sampling without replacement concentrates at least as tightly as with replacement (Serfling, 1974). The total positivity behind the single-crossing step is Karlin (1968); that an order statistic of a dependent sample need not be dispersively smaller than the independent one, which is why we claim only the variance, is Panja et al. (2022); the closest prior comparison of order statistics across dependence, Hu and Hu (1998), orders their *distribution functions* between associated and independent samples rather than the *variance* that is our object; the connection between supermodular ordering and negative dependence is Christofides and Vaggelatou (2004); the Bahadur representation is Bahadur (1966). Determinantal and repulsive designs are Hough et al. (2006); Bardenet and Hardy (2020) and stratification is Cochran (1977); Wieczorek (2023) uses "design-based" conformal prediction in the survey-sampling sense—randomization-based validity under a complex sampling design—distinct from the variance-reduction designs here. The finite signed de Finetti representation is Kerns and

Székely (2006), and the within-dataset, across-x reading of its sign is the companion note (Cotton); the information gap is (Cotton). The de Finetti directing measure behind the positive side is a prior in all but name; the Bayesian–conformal interface, where priors meet finite-sample coverage and the trade is validity against efficiency, is surveyed by Deliu and Liseo (2025), and the dispersion-versus-guarantee tension above is a variance-side instance of it. The contribution here is to make the dependence the control variable for the *variance* of realized coverage: the leading-order fan coefficient, the exact law-of-total-variance decomposition on the positive side, the exact aggregate contraction under negative association (Theorem 4), and, for equicorrelated normals, the exact and dispersively-monotone per-level law. The general single-level negative-association contraction is left as a conjecture. The copula reading (Remark 6) rests on Sklar's theorem (Sklar, 1959; Nelsen, 2006); copulas enter conformal prediction elsewhere through the multivariate *output* (e.g. Sun and Yu, 2024), a different axis from the calibration-set copula here.

Appendix: the quantile-slope reduction for Proposition 3

With $Z_{(k)} = aD + b\eta$, $a = \sqrt{1-\rho}$, $b = \sqrt{(1+(n-1)\rho)/n}$, the budget $(n-1)a^2 + nb^2 = n$ holds and $da/d\rho = -1/(2a)$, $db/d\rho = (n-1)/(2nb)$. Under the natural coupling (the same D, η for all ρ), $\dot{Y} := \partial_\rho Z_{(k)} = -D/(2a) + (n-1)\eta/(2nb)$, and the quantile evolves by $\partial_\rho q_\rho(u) = \mathbb{E}[\dot{Y} \mid Y = q] =: M(q)$. Writing $m(q) = \mathbb{E}[D \mid Y = q]$ and $b\,\mathbb{E}[\eta \mid Y = q] = q - a\,m(q)$,

$$M(q) = -\frac{m(q)}{2a} + \frac{n-1}{2nb^2}\big(q - a\,m(q)\big). \tag{17}$$

Differentiating and using $\frac{1}{2a} + \frac{(n-1)a}{2nb^2} = \frac{nb^2+(n-1)a^2}{2anb^2} = \frac{1}{2ab^2}$ (the budget),

$$M'(q) = \frac{n-1}{2nb^2} - \frac{m'(q)}{2ab^2}, \tag{18}$$

so $M'(q) \geq 0$, dispersive monotonicity ($\partial_\rho q_\rho$ increasing in q), is equivalent to $m'(q) \leq (n-1)a/n$. Tweedie's identity $m'(q) = a\,\mathrm{Var}(D \mid Y = q)/b^2$ turns this into the posterior-variance bound $\mathrm{Var}(D \mid Y = q) \leq (n-1)b^2/n$, which Lemma 2 and the Brascamp–Lieb inequality deliver, as in Proposition 3.

The strong-log-concavity constant in Lemma 2. The constant $n/(n-1)$ is tracked as follows. On the hyperplane $H = \{\sum_i w_i = 0\}$ the centred vector w has $-\log$ density $\frac{1}{2}\|w\|^2$ (the covariance $I - \frac{1}{n}J$ restricts to the identity on H), so the measure is 1-strongly-log-concave there; restricting to the convex cone $\{w_1 \leq \cdots \leq w_n\} \cap H$ adds a convex term to $-\log$ and keeps it 1-strongly-log-concave. Strong log-concavity is preserved under marginalization along a linear functional (Prékopa, 1973; Brascamp and Lieb, 1976; Saumard and Wellner, 2014): for an m-strongly-log-concave measure and $\ell(x) = \langle v, x\rangle$, the law of ℓ is $(m/\|v\|^2)$-strongly-log-concave (the marginal along the unit direction $v/\|v\|$ is m-slc, and rescaling the argument by $\|v\|$ scales the constant by $1/\|v\|^2$). Here $D = w_{(k)} = \langle e_k, w\rangle$ on the cone, with $v = \mathrm{proj}_H(e_k) = e_k - \frac{1}{n}\mathbf{1}$ and $\|v\|^2 = \frac{n-1}{n}$, so $m = 1$ gives the constant $n/(n-1)$ (the cone restriction only raises it). Numerically the realized constant is larger still, $(\log g_D)'' \leq -2.5$ in the cases checked, so the bound holds with room.

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